Pepperdine Live 2.0

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I. Introduction

Pepperdine campus navigation and event awareness remain fragmented across disparate platforms, posing usability challenges for students and visitors. Visitors are frequently found lost on campus having no idea where to go. Students far too often learn about events once they have already happened. To address this, we first drew inspiration from Pepperdine Live, a real-time, interactive data visualization tool created by Hunter Martin for Pepperdine University. The original system visualized campus activity using a 3D map captured via RenderDoc and processed in Blender.

We are pleased to present Pepperdine Live 2.0, an advanced 3D interactive mapping platform designed to elevate the experience of navigating and engaging with the Pepperdine University campus. This project offers a highly detailed three-dimensional terrain model of the campus, complete with accurately rendered and clearly labeled buildings. Each structure is detailed to provide users with a realistic and immersive view of the university's environment.

Pepperdine Live 2.0 also integrates a range of dynamic, real-time features to enhance user interaction and accessibility. Live weather updates are incorporated directly into the platform, ensuring that users are always informed about current campus conditions. In addition, live events are mapped across the campus, complete with comprehensive information regarding event locations, descriptions, and schedules, allowing students, faculty, staff, and visitors to stay connected with campus activities.

The platform includes a powerful search functionality that enables users to locate specific buildings and rooms quickly and efficiently. Floor maps are available for each building, providing a detailed layout of rooms, offices, and facilities throughout the campus. This feature is particularly valuable for new students, guests, and anyone needing to navigate the university with ease and precision.

Pepperdine Live 2.0 represents a significant step forward in campus engagement and information accessibility, combining 3D visualization with real-time data integration. We believe this platform could serve as an essential tool for enhancing the experience of all members of the Pepperdine community.

II. Obtaining model

The initial phase of the Pepperdine Live 2.0 project focused on acquiring and constructing an accurate 3D model of the Pepperdine University campus terrain. To begin, a Google API key was secured, granting access to Google Maps data essential for the development of the terrain model. Utilizing Blender, a standard 3D modeling software, along with a specialized Blender extension known as Bloom, we successfully extracted detailed topographical data from Google Maps and converted it into a usable 3D model.

To create the framework the campus buildings, 3D data from OpenStreetMap was employed. This dataset provided a foundation for buildings and for their initial placement and orientation in the campus landscape. Most importantly, Google Maps 3D Tiles were utilized to create a precise and realistic terrain model of the Pepperdine campus. These tiles offered a highly detailed and geographically accurate representation of the campus.

In parallel, 2D data from OpenStreetMap was also integrated into the project. This data was used to produce a flat, 2D visual representation of the campus, which was later projected onto the 3D terrain model. This method enabled the development of a detailed and layered visualization, combining both accurate elevation modeling and spatial building information, setting a strong foundation for the interactive features that would follow.

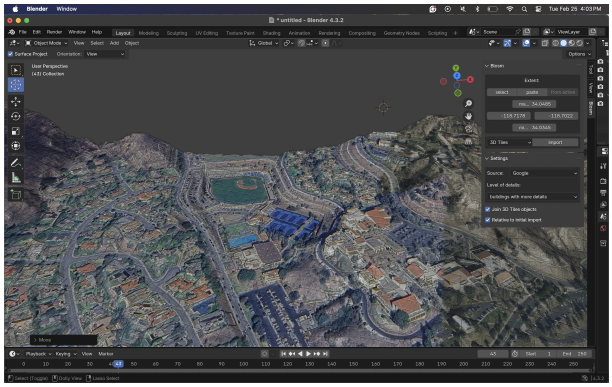


Fig. 1. 3D terrain model

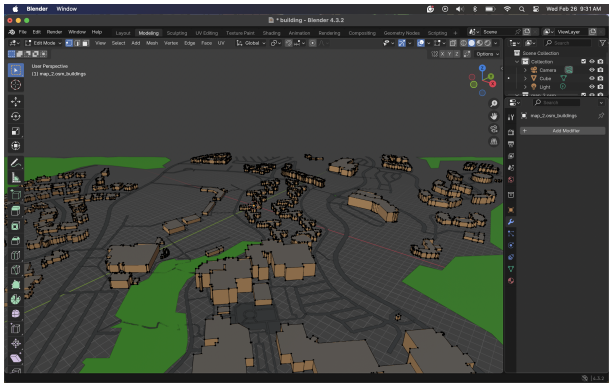


Fig. 2. Building models

Unity Version Control was chosen as the primary development environment for several key reasons. First and foremost, Unity offers robust capabilities for handling 3D models and animations, making it a powerful tool for creating an interactive 3D environment. Moreover, we possessed prior experience with Unity, which contributed to a smoother introduction to the project. Familiarity with the platform's tools and workflows allowed quick progression in the beginning of the project.

Additionally, Unity's versatility and widespread adoption within software development made it a strategic choice. The platform also supports concurrent development, enabling both team members to work simultaneously on different aspects of the project without conflict. Unity Version Control ensured that all changes could be tracked, conflicts could be resolved quickly, and the project's progression remained organized and transparent throughout development.

III. Features

(a) **Live weather integration**
Implemented real-time weather data updates for Pepperdine by writing a script that queries the Open-Meteo API, which is a free and publicy accessible weather data providor. The API provides localized forecasts, which we parsed and displayed through a dedicated UI panel.

(b) **Live Event Integration**
We developed a custom Unity-based XML scraper that collects data from Pepperdine's Trumba event calendar (`https://www.pepperdine.edu/events/`). This script parses RSS feeds, filters for upcoming events, and links them to corresponding campus locations using the Location ID system. Events are displayed in the application with clickable entries that highlight the associated location on the map, if a location is found.

(c) **Campus Building Import and Topological Mapping**
Using 3D buiding models derived from Google Maps, we reconstructed Pepperdine campus within Unity. Buildings were accurately placed to match real-world topology and scale. A key map layer was the topographical map, which allowed us to show Pepperdine as the 3D campus it is in real life.

(d) **Location Identification System**
We designed a Location ID system to index all named locations. Each location entry includes a canonical name, aliases, and optional notes to show to the user. This system serves as a backbone for event tagging, earch functionality, and pathing between locations.

(e) **Pathfinding System**
To ease campus navigation, we implemented a navigation system, using Unity's NavMesh system, to path between known locations. This was done at a stage of the project where only buildings were implemented, and is currently incomplete.

(f) **Search Interface**
We created a search bar UI that supports partial name matches and autofill suggestions. Results from the search connect fo the location ID system, and allows users to focus on a specific room of building, knowing only its name.

(g) **Interior Room Mapping** Populated major campus buildings with interior room data based on official floor plans. Floor plans were imported into Unity, scaled to fit the building outline, and rooms with location ID's were placed in their correct positions. This feature necessitated a building animation system, which, has the system auto-animate occluding sections of a building away so that the user may see the desired room. This floor animation system also allows the user to animate to other floors, to get a better picture of the building as a whole. This enhances map detail and is especially helpful for new students or campus visitors.

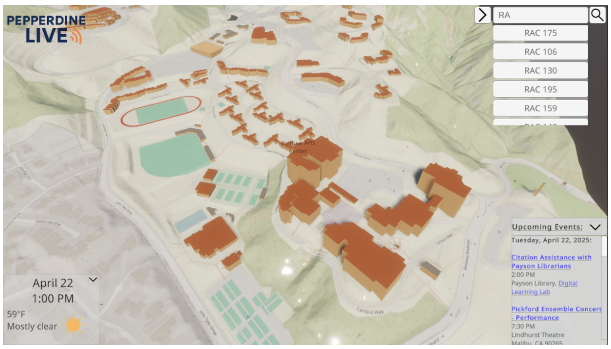


Fig. 3. Floor Plan

IV. Next Steps

While there limitless directions this project can be taken we have identified several first steps that can be used to improve the function, performance, and usability of Pepperdine Live 2.0. These next steps would build upon our project and enable Pepperdine Live 2.0 to be deployed and used across the Pepperdine community.

The biggest focus for future development is the improvement of the pathfinding system to include room navigation. The current pathfinding system routes users between buildings on campus. By extending the pathfinding to navigate inside buildings would raise the utility of the platform. This would allow users to directly path to rooms, offices, or facilities alike, making Pepperdine Live 2.0 an invaluable tool.

Performance optimization is another key step that could be improved on. As we add complexity and further detail to the project performance will need to be taken into consideration. Currently the terrain is made up of over 600,000 faces which most definitely can be reduced. As more building detail and features are added performance must be kept a priority.

To eventually release Pepperdine Live 2.0, we propose chainging the project's rendering pipeline from the High Definition Render Pipeline to the Universal Render Pipeline in Unity. While HDRP offers high level of detail, it does not allow web-based deployment. Converting to URP would allow Pepperdine Live 2.0 to run on the web and on a broad

set of devices. This improves accessibility of the project allowing greater impact.

We would like to improve upon our 3D models used within the platform. Building and room dimensions do not match in scale or height compared to the real campus, which can be confusing. Future improvement would fix these inconsistencies and improving the 3D rendering of the buildings. This would greatly improve the quality of the model and the usability of the navigation.

We also see possible integration of real-time campus activity data to add features that use the Google Places API. Pepperdine Live 2.0 could display live or estimated busy-ness levels across various parts of campus. This feature would give users more information about current events on campus and provide a more responsive experience.

Finally, expanding the interior room mapping remains important. While most campus buildings have been mapped, future development can be made by mapping missed facilities such as the Music Building, the Cultural and Communication Building (CCB), the Graziadio Business School, The Mountain, and other graduate campus facilities. This expansion would ensure functionability to all Pepperdine members and not just the Seaver community.

V. Conclusion

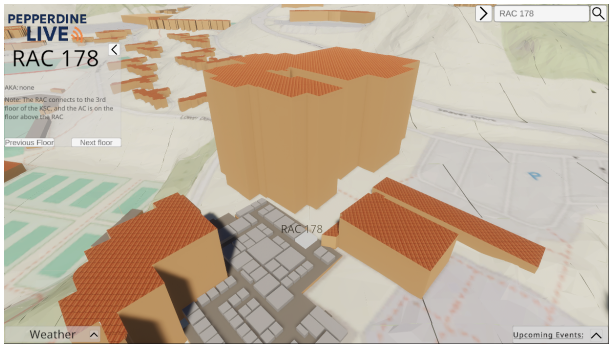


Fig. 4. Final Model

Pepperdine Live 2.0 represents the culmination of our efforts to create an interactive and informative 3D map of the Pepperdine University campus. By integrating features such as live weather updates, active event listings, a search bar for locating buildings and rooms, and detailed layouts of building interiors, our project aims to enhance navigation and accessibility for students, faculty, and visitors alike. Developed in Unity and managed through Unity Version Control, Pepperdine Live 2.0 has the potential to be implemented as an interactive campus directory, accessible via a dedicated website or through digital kiosks placed at key locations across campus. Such integrations would provide real-time guidance, event awareness, and spatial orientation, transforming how the Pepperdine community and its visitors engage with campus spaces.

The full Pepperdine Live 2.0 source files, or build executable files can be found at https://github.com/Hailstorm56435/PepperdineLive2.0